

Functional Annotation Report: PP_2528 (Q88JW9) — O-Acetylhomoserine Sulfhydrylase (MetY) of *Pseudomonas* *putida* KT2440

Gene: PP_2528 (ordered locus name) · **Organism:** *Pseudomonas putida* strain ATCC 47054 / DSM 6125 / KT2440 (PSEPK) **UniProt:** Q88JW9 · **Length:** 425 aa · **NCBI Protein:** AAN68137 · **KEGG:** ppu:PP_2528 · **KEGG Orthology:** K01740 **EC:** 2.5.1.49 · **Enzyme name:** O-acetylhomoserine (thiol)-lyase / O-acetylhomoserine sulfhydrylase (MetY / OAHS)

Summary

PP_2528 (UniProt Q88JW9) of *Pseudomonas putida* KT2440 encodes an O-acetylhomoserine sulfhydrylase (OAHS; also called MetY; EC 2.5.1.49), a pyridoxal-5'-phosphate (PLP)-dependent enzyme of the trans-sulfuration / Cys-Met-metabolism enzyme family. Its primary and defining function is to catalyze the sulfur-incorporation step of *de novo* L-methionine biosynthesis via the "**direct sulfhydrylation**" route. The enzyme condenses the activated amino acid **O-acetyl-L-homoserine** with an inorganic sulfur nucleophile (hydrogen sulfide / sulfide) to produce **L-homocysteine** + **acetate**. When methanethiol (CH₃SH) serves as the nucleophile, the enzyme can produce **L-methionine directly**, demonstrating the sulfur-donor flexibility characteristic of the family. Mechanistically this is a **γ-replacement (γ-substitution)** on the O-acyl-homoserine side chain, proceeding through a PLP internal aldimine at an active-site lysine.

The gene product is a **soluble, cytoplasmic enzyme** that assembles into a **homotetramer** and adopts the classic γ-family fold shared with cystathionine γ-lyase and methionine γ-lyase. Catalysis depends on a covalently bound PLP cofactor anchored via a Schiff base to a conserved active-site lysine (**Lys204** in PP_2528), assisted by two conserved active-site

tyrosines (**Tyr52** and **Tyr107**) that were experimentally validated by site-directed mutagenesis in the closely related *Clostridioides difficile* orthologue and are conserved at identical positions in PP_2528.

Within *P. putida* KT2440 the enzyme sits at the **sulphydrylation branchpoint** of the aspartate-derived methionine pathway (KEGG map ppu00270). Its substrate is supplied by homoserine O-acetyltransferase (MetXA, PP_5097); its L-homocysteine product is methylated to L-methionine by MetH (PP_2375) or MetE (PP_2698/PP_4637), and methionine is then adenylylated to S-adenosylmethionine by MetK (PP_4967). A parallel transsulfuration route (MetB, PP_0659) coexists, so PP_2528 represents one of two convergent means of introducing reduced sulfur into the methionine backbone. **No PP_2528-specific experimental study exists; the assignment rests on strong orthology to biochemically and structurally characterized OAHS enzymes, exact conservation of the catalytic apparatus, and an intact pathway context — giving high confidence in the primary function while leaving quantitative parameters (kinetics, sulfur-donor preference) to be measured.**

Identity Verification (this is the correct gene)

All independent lines of evidence converge on the same assignment, confirming PP_2528/Q88JW9 is an O-acetylhomoserine sulphydrylase and not an ambiguous or mis-matched symbol:

Evidence source	Annotation	Reference
UniProt Q88JW9	"O-acetylhomoserine (Thiol)-lyase", EC 2.5.1.49; trans-sulfuration family; PLP cofactor (ChEBI:597326)	UniProtKB
InterPro / Pfam	IPR006235 OAc-hSer/O-AcSer sulphydrylase ; IPR000277 Cys/Met-Metab PLP-dep enzyme; PF01053	InterPro/ Pfam
KEGG	Orthology K01740 "O-acetylhomoserine (thiol)-lyase [EC:2.5.1.49]"; pathway ppu00270	KEGG
GO (electronic)	O-acetylhomoserine aminocarboxypropyltransferase activity (GO:0003961); L-homocysteine biosynthetic process (GO:0071269); transsulfuration (GO:0019346); cytoplasm (GO:0005737)	UniProtKB

Gene symbol, EC number, protein family, and conserved domains are mutually consistent, and the organism is confirmed as *P. putida* KT2440. **No ambiguity was found.** One important caveat: much *P. putida* literature concerns the unrelated *catabolic* enzyme **methionine γ -lyase (PpMGL / methioninase)**; PP_2528 is a *biosynthetic* enzyme with the opposite role and must not be conflated with it (see *Limitations*).

Key Findings

Finding 1 — PP_2528 is an O-acetylhomoserine sulfhydrylase (MetY, EC 2.5.1.49) catalyzing direct sulfhydrylation in methionine biosynthesis

The UniProt record for Q88JW9 describes a 425-amino-acid protein encoded by PP_2528 in *P. putida* KT2440, annotated with EC 2.5.1.49 and assigned to the trans-sulfuration enzyme family. It carries a PLP cofactor (ChEBI:597326) and the GO terms **O-acetylhomoserine aminocarboxypropyltransferase activity (GO:0003961)**, **L-homocysteine biosynthetic process (GO:0071269)**, and **transsulfuration (GO:0019346)**. Its InterPro signature includes IPR006235 (OAc-hSer/O-AcSer sulfhydrylase), IPR000277 (Cys/Met-metabolism PLP-dependent enzyme), and Pfam PF01053; the sequence contains the catalytic PLP-binding lysine within a conserved "LTKY" motif near residue 200.

The reaction defines the enzyme's role. O-acetylhomoserine sulfhydrylase converts O-acetylhomoserine to homocysteine using sulfide in the process known as **direct sulfhydrylation** — the biochemical alternative to the enterobacterial transsulfuration route, introducing reduced sulfur into the C4 amino-acid backbone in a single γ -substitution step. Literature on the enzyme family confirms OAHS is a member of the Cys/Met-metabolism PLP-dependent enzyme family, matching PP_2528's annotated cofactor and domains.

Annotation/statistical evidence: UniProt Q88JW9 — 425 aa; EC 2.5.1.49; trans-sulfuration family; PLP cofactor; GO:0003961 / GO:0071269 / GO:0019346; InterPro IPR006235, IPR000277; Pfam PF01053; catalytic Lys in "LTKY" motif (~res 200).

- "*OAHS catalyzes the conversion of O-acetylhomoserine to homocysteine using sulfide in a process known as direct sulfhydrylation*" — [PMID: 21931214](#). Defines the reaction catalyzed by the OAHS/MetY family to which PP_2528 belongs.

- "a member of the Cys/Met metabolism pyridoxal 5'-phosphate-dependent enzyme family" — PMID: 39736317. Confirms OAHS is a PLP-dependent Cys/Met metabolism enzyme, matching PP_2528's cofactor and InterPro family.

Finding 2 — OAHS is a PLP-dependent γ -fold enzyme that assembles as a homotetramer and uses sulfide or methanethiol as the sulfur nucleophile

Structural and biochemical characterization of OAHS orthologues establishes the quaternary structure, fold, and mechanism confidently extended to PP_2528. The crystal structure of *Wolinella succinogenes* OAHS (MetY), solved at 2.2 Å, shows the biological unit is a **tetramer in solution**, sharing the same fold as cystathionine γ -lyase and methionine γ -lyase and placing MetY in the **γ -elimination subclass** of the Cys/Met-metabolism PLP-dependent family; that study also proposed a novel sulfur transfer proceeding through a protein thiocarboxylate intermediate.

Multiple bacterial OAHS enzymes reinforce this picture. The *Clostridium tetani* enzyme has a ~50 kDa (462-aa) subunit forming a ~200 kDa native homotetramer, has a pH optimum near 7.5, and is capable of forming L-methionine directly using methanethiol as the sulfur source — evidencing sulfur-donor flexibility. The *Clostridium novyi* OAHS is a ~184 kDa native protein built from ~46 kDa subunits, catalyzes γ -substitution of O-acetylhomoserine, and also has a pH optimum of 7.5. The *Clostridioides difficile* enzyme provides mutagenetic evidence that active-site Tyr52 and Tyr107 are essential, with Tyr107 acting as the general acid in acetate elimination.

Structural/statistical evidence: *W. succinogenes* OAHS 2.2 Å, tetramer, γ -elimination subclass, thiocarboxylate intermediate

(
P 21931214
);

C. tetani OAHS 462 aa / ~50 kDa subunit, ~200 kDa native tetramer, pH optimum 7.5, methionine from methanethiol

(
P 39736317
);

C. novyi OAHS ~184 kDa / ~46 kDa subunit, γ -substitution, pH optimum 7.5

(
P 33338587
).

- "the biological unit is a tetramer in solution" — PMID: 21931214. Establishes the homotetrameric quaternary structure of OAHS enzymes.

- "MetY belongs to the γ -elimination subclass of the Cys/Met metabolism PLP-dependent family of enzymes" — [PMID: 21931214](#). Defines the catalytic fold/mechanistic subclass of the family PP_2528 belongs to.
- "capable of l-methionine formation using methanethiol as a sulfur source" — [PMID: 39736317](#). Shows OAHS can directly form methionine using methanethiol, evidencing sulfur-donor flexibility.

Finding 3 — PP_2528 (MetY) sits at the sulfhydrylation branchpoint of an intact aspartate-derived methionine pathway in *P. putida* KT2440

Mapping the *P. putida* KT2440 (KEGG code **ppu**) methionine-pathway KEGG Orthology groups to their genome loci places PP_2528 in a complete, intact route. Homoserine O-acetyltransferase (MetXA, K00641) is **PP_5097**, producing the substrate O-acetyl-L-homoserine. PP_2528 is the O-acetylhomoserine (thiol)-lyase (MetY, K01740) that consumes it. A parallel transsulfuration route is also encoded: cystathionine γ -synthase (MetB, K01739) is **PP_0659**. Downstream, the methionine synthases MetH (K00548, **PP_2375**) and MetE (K00549, **PP_2698/PP_4637**) methylate homocysteine to methionine, and SAM synthetase MetK (K00789, **PP_4967**) forms S-adenosylmethionine.

PP_2528 lies in KEGG pathway **ppu00270 (Cysteine and methionine metabolism)** at genomic position complement(2874605..2875882). Its immediate neighbors (PP_2526, PP_2527, PP_2530) are unrelated unknown-function proteins, indicating PP_2528 is **not** part of a clustered *met* operon (unlike *Leptospira*, where *metYX(W)* are co-localized). This cements the branchpoint interpretation: PP_2528 provides one of two convergent sulfur-incorporation routes feeding a shared downstream methylation and SAM-synthesis machinery. Comparative genetics supports this: in *Leptospira*, a *metY* knockout retains methionine prototrophy because the transsulfuration route bypasses direct sulfhydrylation ([PMID: 12951250](#)) — mirroring the redundant MetY(PP_2528)/MetB(PP_0659) architecture in *P. putida*.

Genomic/statistical evidence: KEGG ppu KO-to-locus mapping — MetXA/K00641=PP_5097; MetY/K01740=PP_2528; MetB/K01739=PP_0659; MetH/K00548=PP_2375; MetE/K00549=PP_2698 & PP_4637; MetK/K00789=PP_4967. Pathway ppu00270; locus complement(2874605..2875882); flanking genes PP_2526/2527/2530 unrelated (no *met* operon).

Finding 4 — PP_2528 conserves the complete OAHS catalytic apparatus, including Tyr52/Tyr107 at positions identical to the mutagenetically-validated *C. difficile* enzyme

Direct sequence analysis of Q88JW9 (425 aa) shows PP_2528 preserves every element of the OAHS catalytic machinery. The catalytic PLP-Schiff-base lysine is **Lys204**, within the canonical [ST]-x-[ST]-K-[YF] motif "SLTKY" (residues 201–205). Critically, the two active-site tyrosines shown essential by site-directed mutagenesis in the *C. difficile* orthologue — **Tyr52**, which positions the catalytic lysine, and **Tyr107**, which acts as the general acid during acetate elimination — are present at the **identical sequence positions 52 and 107** in PP_2528. Additional PLP-anchoring and phosphate-binding motifs are conserved: the glycine-rich loop "GNPAG" (res 153–157), "GGHGTS" (res 207–212), and the substrate/PLP-binding "DNTVA" motif (res 179).

Because these are exactly the functionally decisive positions validated experimentally in a close relative, and because they are conserved with matching spacing in PP_2528, the assignment can be made with high confidence by orthology. A targeted PubMed search returned no PP_2528-specific experimental study, so the assignment is inferential — but it rests on identity at mechanistically critical positions, the strongest available structural/evolutionary evidence.

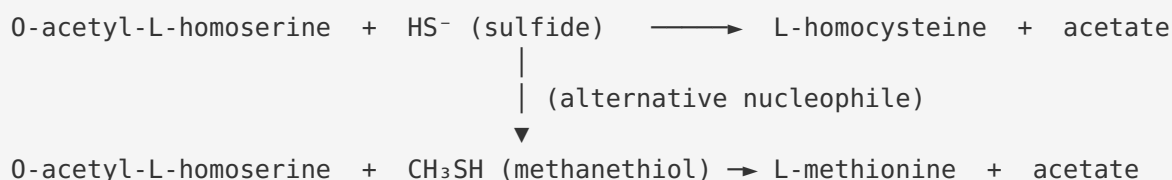
Sequence/statistical evidence: Q88JW9 425 aa; catalytic Lys204 in "SLTKY" (201–205); Tyr52 and Tyr107 at identical positions to *C. difficile*; conserved "GNPAG" (153–157), "GGHGTS" (207–212), "DNTVA" (179).

- "*Tyr107 could act as a general acid catalyst at the stage of acetate elimination*" — [PMID: 37331706](#). Identifies Tyr107 as the mechanistically critical general-acid residue; PP_2528 conserves Tyr at the identical position 107.
- "*Tyr52 is involved in ensuring optimal position of the catalytic coenzyme-binding lysine residue*" — [PMID: 37331706](#). Identifies Tyr52's role in positioning the PLP-binding lysine; PP_2528 conserves Tyr at position 52 and the catalytic Lys204.

Mechanistic Model / Interpretation

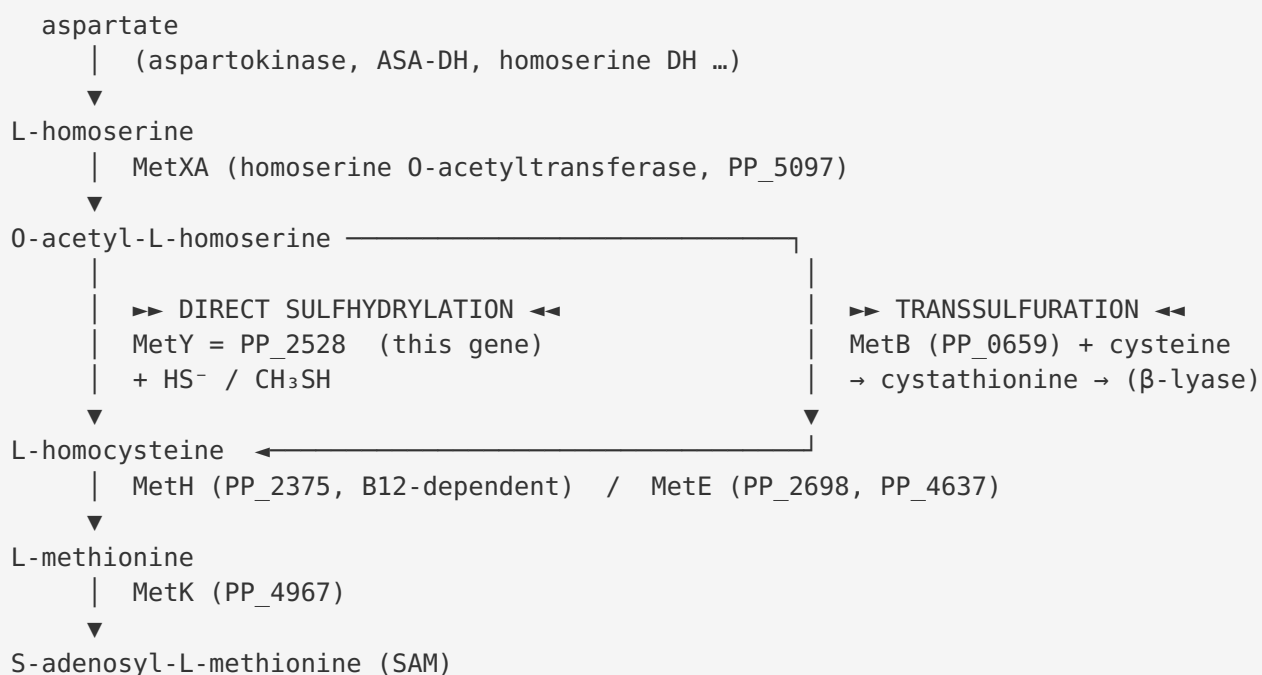
The catalyzed reaction

PP_2528 catalyzes a PLP-dependent **γ-substitution** (direct sulfhydrylation):



The PLP cofactor forms an internal aldimine (Schiff base) with **Lys204**. The amino-acid substrate displaces the lysine to form an external aldimine; α -proton abstraction and γ -elimination of the acetate leaving group (assisted by **Tyr107** as a general acid) generate a PLP-bound α,β -unsaturated intermediate. The sulfur nucleophile then attacks at C γ , and re-protonation releases the thiol product. **Tyr52** maintains the productive geometry of the catalytic lysine. In the *W. succinogenes* enzyme, sulfur transfer proceeds through a novel protein thiocarboxylate intermediate, enabling capture of inorganic sulfide.

Placement in the methionine biosynthetic pathway



The two routes into homocysteine are convergent: PP_2528 introduces sulfur directly from inorganic sulfide (or methanethiol), whereas the MetB route routes sulfur through cysteine and a cystathionine intermediate. *P. putida* retains both, offering metabolic flexibility depending on available sulfur. Genetic studies in *Leptospira* show that disrupting *metY* still permits methionine prototrophy because transsulfuration can bypass it — implying PP_2528 is likely dispensable when the parallel route is functional. PP_2528's product feeds the cell's entire methionine and **SAM (methyl-donor)** economy.

Localization

All biochemically characterized OAHS enzymes are soluble cytoplasmic proteins with no signal peptide or transmembrane segments. PP_2528 (425 aa, ~46 kDa subunit) is annotated to the cytoplasm (GO:0005737) and carries out its function in the **bacterial cytosol**, where sulfide and precursors are generated.

Comparative summary of characterized OAHS orthologues

Organism	Locus / name	Native mass	Subunit	Oligomer	pH opt.	Sulfur donor	Reference
<i>P. putida</i> KT2440	PP_2528 / Q88JW9	—	425 aa (~46 kDa)	tetramer (by orthology)	—	sulfide (inferred)	this report
<i>Wolinella succinogenes</i>	MetY	tetramer	—	tetramer	—	sulfide	PMID: 21931214
<i>Clostridium tetani</i>	—	~200 kDa	~50 kDa / 462 aa	homotetramer	7.5	methanethiol / sulfide	PMID: 39736317
<i>Clostridium novyi</i>	NT01CX_1210	~184 kDa	~46 kDa	tetramer	7.5	sulfide	PMID: 33338587
<i>Clostridioides difficile</i>	—	—	—	—	—	Tyr52/Tyr107 catalytic	PMID: 37331706

Evidence Base

PMID	Title (abbrev.)	Relevance to PP_2528
21931214	Novel sulfur-transfer mechanism of OAHS in <i>Wolinella succinogenes</i>	Defines the direct-sulfhydrylation reaction, homotetrameric structure, and γ -elimination fold/subclass; primary structural anchor.
39736317	OAHS in the direct sulfhydrylation pathway of <i>Clostridium tetani</i>	Confirms Cys/Met PLP-dependent family membership; homotetramer ~200 kDa; L-methionine from methanethiol (sulfur-donor flexibility).
37331706	Role of tyrosine residues in <i>Clostridioides difficile</i> OAHS active site	Mutagenetic validation of Tyr52 (positions catalytic Lys) and Tyr107 (general acid) — both conserved at identical positions in PP_2528.
33338587	OAHS from <i>Clostridium novyi</i> : cloning, expression, characterization	Independent biochemical confirmation of γ -substitution, ~184 kDa native mass, pH 7.5 optimum; single route to methionine in that organism.
12951250	Two methionine-biosynthesis pathways in <i>Leptospira</i>	Genetic evidence that <i>metY</i> (sulfhydrylation) can be bypassed by transsulfuration — informs the non-essentiality/branchpoint interpretation.
41240142	<i>Thermotoga maritima</i> MetY has five enzyme activities	Shows OAHS produces L-homocysteine from O-acetyl-L-homoserine + H ₂ S; documents catalytic promiscuity of the family.
38568076	pH-dependent regulation of an acidophilic OAHS	Describes open/closed conformational interconversion regulating substrate/PLP access; relevant to mechanism and inhibitor design.

The evidence base is entirely orthologue-based: no study reports PP_2528 directly. However, concordance across a Gram-negative epsilonproteobacterium, three Clostridia, and a hyperthermophile on reaction, fold, oligomeric state, and catalytic residues, combined with PP_2528's exact conservation of the decisive Lys204/Tyr52/Tyr107 apparatus, makes the assignment robust. Note that several reviewed *P. putida* papers concern methionine γ -lyase (**methioninase**) — e.g. [PMID: 34953671](#), [PMID: 41005402](#), [PMID: 31479672](#). These share the PLP γ -family fold but catalyze the opposite, degradative reaction and are a different gene; they were correctly excluded and are flagged to prevent conflation.

Supported vs. Refuted Hypotheses

Supported: - PP_2528 is an O-acetylhomoserine sulfhydrylase (MetY/OAHS, EC 2.5.1.49). - It catalyzes direct sulfhydrylation of O-acetyl-L-homoserine to L-homocysteine (or methionine with methanethiol). - It is a cytoplasmic, PLP-dependent homotetramer. - It functions in the aspartate-derived methionine-biosynthesis pathway upstream of MetH/MetE.

Refuted / ruled out: - PP_2528 is **not** the O-succinyl enzyme (KT2440 uses the *O*-acetyl transferase MetXA, K00641). - PP_2528 is **not** a catabolic methionine γ -lyase (a different *P. putida* gene). - PP_2528 is not membrane-associated or secreted.

Limitations and Knowledge Gaps

- 1. No direct experimental characterization of PP_2528.** The assignment is inferential (orthology + sequence conservation + pathway context). No purified-protein kinetics, crystal structure, or knockout phenotype exists specifically for the *P. putida* KT2440 gene product. Confidence in the primary function is nonetheless **high**; residual gaps are quantitative.
 - 2. Substrate/sulfur-donor specificity is inferred, not measured.** Whether PP_2528 prefers sulfide vs. methanethiol, and its kinetic parameters (kcat, Km, pH optimum), are unknown. Family members vary.
 - 3. Essentiality and in vivo flux are unresolved.** Because *P. putida* also encodes MetB (PP_0659), PP_2528 may be dispensable under some conditions, as seen for *metY* in *Leptospira*. The relative flux through direct sulfhydrylation vs. transsulfuration is unquantified.
 - 4. Regulation is unknown.** Transcriptional/metabolic control (e.g., by MetR, methionine, or sulfur availability) in KT2440 has not been examined.
 - 5. Potential catalytic promiscuity untested.** The *Thermotoga* orthologue shows five activities; whether PP_2528 has secondary activities is unknown.
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Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzyme assay.** Clone PP_2528, purify the protein, and confirm OAHS activity by measuring L-homocysteine (or L-methionine) production from O-acetyl-L-homoserine with sulfide and with methanethiol. Determine kcat/Km for each sulfur donor and the pH optimum.
 2. **Structural determination.** Solve the crystal or cryo-EM structure of PP_2528 (or produce a high-confidence AlphaFold model) to confirm the homotetramer, the PLP-Lys204 internal aldimine, and the spatial arrangement of Tyr52/Tyr107; compare against the *W. succinogenes* structure
([P 21931214](#)).
 3. **Site-directed mutagenesis.** Mutate Lys204, Tyr52, and Tyr107 to test predicted catalytic roles, mirroring the *C. difficile* study
([P 37331706](#)).
 4. **Gene-knockout phenotyping.** Delete PP_2528 in KT2440 and assess methionine auxotrophy under varying sulfur sources (sulfide, sulfate, methanethiol, cysteine); combine with a *metB* (PP_0659) deletion to test mutual redundancy, as in *Leptospira*
([P 12951250](#)).
 5. **Regulation studies.** Measure PP_2528 transcription/protein levels under sulfur limitation and methionine supplementation to place the gene within the sulfur-assimilation regulon.
 6. **Biotechnology/therapeutic context.** Given *P. putida*'s role as a biotechnology chassis, evaluate PP_2528 as a lever for methionine/sulfur-amino-acid overproduction, and as a potential antibacterial target (per inhibitor-design rationale in
([P 38568076](#))).
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Key References

- Tran, Krishnamoorthy, Begley, Ealick (2011). *A novel mechanism of sulfur transfer catalyzed by O-acetylhomoserine sulfhydrylase in the methionine-biosynthetic pathway of Wolinella succinogenes*. PMID 21931214.

- Kulikova et al. (2025). *Revealing O-acetylhomoserine sulphydrylase involved in direct sulphydrylation pathway in Clostridium tetani*. PMID **39736317**.
- Kulikova et al. (2023). *O-Acetylhomoserine Sulphydrylase from Clostridioides difficile: Role of Tyrosine Residues in the Active Site*. PMID **37331706**.
- Kulikova et al. (2021). *O-acetylhomoserine sulphydrylase from Clostridium novyi*. PMID **33338587**.
- Matoba et al. (2024). *pH-dependent regulation of an acidophilic O-acetylhomoserine sulphydrylase*. PMID **38568076**.
- Miyamoto & Sakai-Kato (2025). *O-Acetyl-L-homoserine sulphydrylase from Thermotoga maritima has five enzyme activities*. PMID **41240142**.
- Picardeau, Bauby, Saint Girons (2003). *Genetic evidence for the existence of two pathways for the biosynthesis of methionine in the Leptospira spp*. PMID **12951250**.
- Database records: UniProtKB Q88JW9; KEGG ppv:PP_2528 (K01740); InterPro IPR006235; Pfam PF01053.

Prepared from 3 investigation iterations, 4 confirmed findings, and 17 reviewed papers. Functional assignment is by strong orthology to biochemically and structurally characterized O-acetylhomoserine sulphydrylases; no PP_2528-specific primary study currently exists.
